MIL-D-45441D (PA)
5 April 1974
SUPERSEDING
MIL-D-45441C (MU)
1 June 1966 and
MIL-D-45493C (MU)
1 June 1966
(See 6.6)

MILITARY SPECIFICATION

DETONATOR, STAB, M61
PARTS FOR
AND LOADING, ASSEMBLING AND PACKING

*This Specification is approved for use by all Departments and Agencies of the Department of Defense.

*1. SCOPE

1.1 This specification contains quality assurance provisions and special requirements for the parts, loading, assembling and packing of one type of initiating device designated as Detonator, Stab, M61.

*2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-A-2550 - Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1375

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-650 - Explosives, Sampling, Inspection and Testing
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT AND PACKAGING DRAWINGS

9258182	- Box, Set Up, Packing, Ammunition, Inner for Small Explosive Components
9258183	- Box, Set Up, Packing, Ammunition, Outer for Small Explosive Components
9258184	- Box, Packing, Ammunition for Small Explosive Components
8865554	- Detonator, Stab, M61, Assembly

INSPECTION EQUIPMENT DRAWINGS

8816367	- Guide, Firing Pin
8827828	- Sleeve
8855295	- Disc, Lead-Coined
8868629	- Ball, Steel
8868631	- Assembly, Firing
9200012	- Pin, Firing
9200602	- Bushing

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*3. REQUIREMENTS

8.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Detonator.-The detonator shall comply with all requirements specified on Drawing (dwg.) 8865554 and with all requirements specified in applicable specifications.

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3.3 Moisture content

3.3.1 Lead azide.-The moisture content of the lead azide used in loading the detonator shall not exceed 0.50 percent.

3.3.2 Primer mixture.-The moisture content of the primer mixture used in loading the detonator shall not exceed 0.30 percent.

3.4 Functioning

3.4.1 Output.-The detonators shall function, on the first attempt under an applied energy of 4.95 inch ounces, and shall cause a .400 inch diameter perforation in a .065 inch min. thick coined lead disc.

3.4.2 Waterproofness.-The detonators shall be waterproof as evidenced by normal functioning, as specified in 3.5.1 after being immersed in water maintained at 75 degrees Fahrenheit (F.) plus or minus 5 degrees F., at a depth of 2 inches minimum for not less than 48 hours.

3.5 Workmanship.-All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

3.6 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

*4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection re-

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quirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First Article Inspection

4.2.1 Submission. - Prior to the start of regular production, the contractor shall submit a first article sample (see 6.1c) to a Government approved facitity as designated by the contracting officer for evaluation in accordance with the provisions of 4.2.2 and 4.2.3. The first article sample shall consist of three hundred forty (340) loaded primers and 50 complete sets of parts (i.e., 50 each of every component), which have been produced by the contractor or furnished by a

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supplier and which have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract. All parts and material, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample, including a certificate of conformance for materials. A sample containing know defects will not be submitted unless specifically authorized by the Contracting Officer. A first article sample, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing process, material used, drawing or specfication such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Components will be subjected by the Government to any or all of the examinations and tests specified in this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection.-If any component fails to comply with the applicable requirements, the first article sample shall be rejected. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production unless otherwise directed by the contracting officer.

4.2.4 Loaded primers. - A sample of three hundred forty (340) loaded primers shall be submitted in accordance with 4.2.1 and tested as specified in 4.3.3.3.

*4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is

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drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded detonator assemblies shall contain:

a. Parts from one lot interfix number from one manufacturer.

b. Primer mixture from one batch or blending of batches which results in homogeneity of all primer mixtures used in loading the entire detonator lot.

c. Lead azide of the same type from not more than two lots produced by the same manufacturer in accordance with the same specification and same specification revision and which have subsequently been subjected to uniformly controlled drying and operating conditions at the loading plant.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to

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all of the major and minor defects listed in the applicable subparagraphs of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4

4.3.2.1 Disc, detonator (see dwg. 8865557 covering a detail of dwg. 8865554).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Diameter	Gage	01001
102.	Thickness	Gage	01002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship, (see 3.5)	Visual	01003

4.3.2.2 Disc, closing (see dwg. 8865556 covering a detail of dwg. 8865554).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Diameter	Gage	02001
102.	Thickness	Gage	02002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	02003

4.3.2.3 Cup, detonator (see dwg. 8865555 covering a detail of dwg. 8865554).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		

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Major:	AQL 0.65 percent		
101.	Diameter of hole, min	Gage	03001
102.	Concentricity of flash hole with outside diameter	Gage	03002
103.	Thickness through bottom	Gage	03003
Minor:	AQL 1.50 percent		
201.	Length	Gage	03004
202.	Outside diameter	Gage	03005
203.	Inside diameter	Gage	03006
204.	Evidence of poor workmanship (see 3.5)	Visual	03007

4.3.2.4 Assembly, prior to loading (see dwg. 8865554).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Coating compound or lacquer on exposed surface of disc	Visual	04001
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.5)	Visual	04002

4.3.2.5 Assembly (see dwg. 8865554).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup cracked or split	Visual	05001
Major:	AQL 1.00 percent		
101.	Length	Gage	05002
102.	Diameter	Gage	05003
103.	Disc loose, punctured, torn or wrinkled to such a degree that the contents are exposed	Visual	05004
104.	Color of lacquer incorrect	Visual	05005
105.	Cup damaged except as described in critical above	Visual	05006
106.	Crimp not full 360 degrees	Visual	05007

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship, (see 3.5).....	Visual	05008

4.3.2.6 Box, set-up, inner, after loading and prior to closing slide cover (see dwg. 9258182).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Filler and cushion assembly missing.....	Visual	06001
2.	Filler and cushion assembly improperly assembled.....	Visual	06002
3.	Stiffener missing.....	Visual	06003
4.	Slide cover assembled reversed.....	Visual	06004
Major:	AQL 0.40 percent		
101.	Release paper missing.....	Visual	06005
Minor:	AQL 1.00 percent		
201.	Marking missing, incomplete or illegible.....	Visual	06006
202.	Evidence of poor workmanship, (see 3.5).....	Visual	06007

4.3.2.7 Box, set-up, outer, with inner set-up boxes assembled and prior to assembling top and sealing strip (see dwg. 9258183).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up box.....	Visual	07001
2.	Stiffener not facing one direction.	Visual	07002
Major:	None defined.		
Minor:	AQL 1.00 percent		
201.	Marking missing, incomplete or illegible.....	Visual	07003
202.	Evidence of poor workmanship, (see 3.5).....	Visual	07004

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4.3.2.8 Box, packing, ammunition prior to assembling cover and strapping (see dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Bags containing set-up boxes damaged	Visual	08001
102.	Cushioning material incomplete or missing from around outer set-up boxes	Visual	08002
103.	Marking missing, misleading or unidentified	Visual	08003
104.	Barrier bag seal incomplete or improperly positioned	Visual	08004
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship, (see 3.5)	Visual	08005

4.3.2.9 Box, packing, ammunition (see dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Strapping missing, broken, loose or improperly fastened	Visual/ Manual	09001
102.	Box damaged	Visual	09002
103.	Marking missing, incomplete or illegible	Visual	09003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship, (see 3.5)	Visual	09004

*4.3.3 Testing

4.3.3.1 Moisture content of lead azide (see 3.3) - Major defect - Code No. 05009.-The moisture content of the lead azide shall be determined for each lot at the time of loading. A

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sample in sufficient quantity to perform the test detailed in 4.4.1 shall be selected and tested as specified in 4.4.1. If the moisture content is in excess of the percentage specified and loading has not begun the lot of lead azide shall be rejected. If loading has begun, action on the unloaded lead azide shall be as specified above and all detonators loaded with the questionable lead azide shall be rejected.

4.3.3.2 Primer Mixture

4.3.3.2.1 Moisture content of primer mixture (see 3.4) - Major defect - Code No. 05010.-The moisture content of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample, in sufficient quantity to perform the test detailed in 4.4.2, shall be selected and tested as specified in 4.4.2. If the moisture content is in excess of the percentage specified and loading has not begun, the batch or blend of batches of primer mixture shall be rejected. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

4.3.3.2.2 Composition (see dwg. 8865554) - Major defect - Code No. 05011.-The composition of the primer mixture shall be determined for each batch or blend of batches at the time of loading. A sample, in sufficient quantity to perform the test detailed in 4.4.2, shall be selected and tested as specified in 4.4.2. If the composition of the primer mixture fails to conform to the requirements specified on the applicable drawing and if loading has not begun, the batch or blend of batches of primer mixture shall be rejected. If loading has begun, action on the unloaded primer mixture shall be as specified above and all detonators loaded with the questionable primer mixture shall be rejected.

4.3.3.3 Functioning

4.3.3.3.1 Output (see 3.5.1) - Major defect - Code No. 05012.-The sample size, acceptance or rejection of the lot shall be in accordance with Table I. The test shall be performed as specified in 4.4.3.1 using test equipment in accordance with 4.3.4 and 4.4.3.1. If more than one loading machine is used in the production of the inspection lot, the sample shall be selected so as to represent proportional quantities of production from each machine.

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TABLE I

Lot Size	Sample	Accept	Reject
to 150,000	First 315	0	3
	Second 315	3	4
150,001 to 300,000	First 500	1	4
	Second 500	4	5
300,001 to 500,000	First 800	2	5
	Second 800	6	7
500,001 and over	First 1250	3	7
	Second 1250	8	9

4.3.3.3.2 Waterproofness (see 3.5.2) Major defect - Code No. 05013.-A sample of 25 detonators shall be selected at random from each lot. If one or more detonators fail to comply with the requirement specified the lot shall be rejected. The test shall be performed as specified in 4.4.3.2 using test equipment as specified in 4.3.4 and 4.4.3.1.

4.3.3.4 Heat seal test of sealed bag (see dwg. 9258183) - Major defect - Code No. 07005.-Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL-P-116.

4.3.4 Inspection equipment.-Inspection equipment required to perform the inspection prescribed in this specification is identified in the Examination (4.3.2) and Test Method (4.4) paragraphs herein. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

*4.4 Test methods and procedures

4.4.1 Moisture content of lead azide and primer mixture.- The moisture content of the lead azide and primer mixture shall be determined in accordance with MIL-STD-650, Method 101.6 except that a two (2) gram sample shall be used.

4.4.2 Composition of primer mixture.-The composition of the primer mixture shall be determined by performing the following tests in the sequence listed.

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4.4.2.1 Barium nitrate.-An accurately weighed portion of approximately 0.5 gram of primer mixture shall be transferred to a tared medium-porosity, fritted glass crucible. Three ml of lead azide saturated distilled water shall be added at a temperature of 50 plus or minus 2 degrees C., agitated for 1 minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure, shall be repeated five times, making a total of six 3-ml. extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, aspirate until the odor of alcohol is no longer detectable, dried in a oven maintained at 60 plus or minus 5 degrees C., for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate in the sample on a dry basis as follows:

$$\text{Barrium nitrate, percent} = \frac{A-B}{W}$$

Where:

A = weight of crucible and sample
B = weight of the crucible and sample after extractions with lead azide saturated water
W = weight of sample

4.4.2.2 Lead styphnate

4.4.2.2.1 Basic lead styphnate (when applicable).-The residue obtained in the determination of barium nitrate shall be extracted with six-3ml. portions of saturated ammonium acetate solution at about 25 degrees C., (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250-ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2-ml. aliquot of this solution shall be transferred to a 50-ml. calibrated volumetric flask and diluted to the mark with distilled water. A portion of this selection shall be transferred to a corex glass spectrophotometric cell having a width of approximately 1 centimeter (cm.), and its absorbance density determined as a wave length of 410 millimicrons. As reference,

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a similar cell containing ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the absorbance density shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the absorbance density of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated as follows:

Basic lead styphnate, percent = F(A-B) divided by W(1-.01M)D

Where:

A = absorbance density of basic lead styphnate solution
B = absorbance density of ammonium acetate solution
D = width of corex cell, cm.
M = moisture content of sample obtained from 4.4.1 in percent
W = weight of sample.
F = Factor to be determined for each instrument on the basis of a sample of lead styphnate that is suitable for use as a standard.

4.4.2.2.2 Determination of normal lead styphnate (when applicable). Use method 408.1 of MIL-STD-650.

4.4.2.3 Tetracene.-The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol aspirated until the odor of alcohol is gone, then dried in an oven maintained at 60 plus or minus 5 degrees C. for 30 minutes, colled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125-ml. beaker with a stream of water, 25 ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible and the residue in the crucible washed three times with boiling water, and finally with alcohol, then aspirate until the odor of alcohol is no longer detectable. The crucible and contents shall be dried in an oven maintained at 60 plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample on a dry basis as follows:

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Tetracene, percent = (C-D) 100 divided by W (1-.01M)

Where:

C = weight of crucible and residue after extration with saturated ammonium acetate solution
D = weight of crucible and residue after treating with hot water
W = weight of the sample
M = moisture content of sample obtained from 4.4.1 in percent

4.4.2.4 Antimony sulfide.-The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the sample on a dry basis as follows:

Antimony sulfide, percent = (D-E) 100 divided by W(1.-.01M)

Where:

D = weight of crucible and sample after treating with hot water
E = weight of crucible
W = weight of sample
M = moisture content of sample obtained in 4.4.1 in percent

*4.4.3 Functioning

4.4.3.1 Output.-The detonator shall be assembled to the test fixture so that the firing pin will strike the sensitive end of the detonator. A steel ball 3/4 inchs in diameter and weighing .99 ounces maximum, shall be dropped on the firing pin to function the detonator, from a height of 5 inches, maximum. The height of the drop shall be measured as the height from the top of the firing pin to the bottom of the ball. A detonator that fails to fire or fails to create a hole in the coined lead disc of the required size shall be classed defective. Method of inspection shall be in accordance with dwgs. 8816367, 8827828, 8855295, 8868629, 8868631, 9200012 and 9200602.

4.4.3.2 Waterproofness.-The detonator shall be immersed in water to the depth and time specified. Within one hour after removal from the water, the detonator shall be subjected to the sensitivity test specified in 4.4.3.1. Any detonator which fails to fire or fails to create a hole in the lead disc of the required size, shall be classed defective.

*5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

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5.1.1 Level C.-Packaging shall be as specified on dwgs. 9258182 and 9258183.

5.2 Packing

5.2.1 Level C.-Packing shall be as specified on dwg. 9258184.

5.3 Marking.-Marking of all packaging and packing shall comply with dwgs. 9258182, 9258183 and 9258184.

*6. Notes

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Euquipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

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b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists (DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Check test for possible deterioration.-The following test is designed to prevent the loading of detonators which have deteriorated, into a major ammunition item. The test shall be

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performed on detonators from each lot which has been stored under normal conditions for more than two years or adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are loaded does not require that the check test be performed, the contracting office should insert a provision for the performance of the check test in that contract. The contracting officer will arrange for the contractor to be reimbursed for expenses incurred in the performance of the check test for possible deterioration.

6.4.1 Functioning.-A sample of 315 detonators shall be selected at random from each lot. If 4 or more detonators fail to comply with the requirement specified, the lot shall be rejected. If more than one defective is found, a second sample of 315 shall be selected at random and tested. If in the combined first and second sample a total of 5 or more defectives are found the lot shall be rejected. The test shall be performed as specified in 4.4.3.1 using test equipment as specified in 4.3.4 and 4.4.3.1.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convinience only and the Government assumes no liability whatsoever for any inadequacies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Supersession data.-This specification includes the requirements of MIL-D-45439C (MU), dated 1 June 1966.

Custodian:
Army-PA

Preparing Activity:
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